

Atopic dermatitis (AD) and contact dermatitis (CD) are two types of inflammatory skin diseases that differ in key characteristics. AD is a chronic condition that commonly affects the skin covering several joints, both exposed to environmental allergens as well as covered by clothing, whereas CD is usually confined to skin that comes into contact with allergens or irritants. AD typically involves the epidermal layer of the skin whereas CD may involve both the epidermis and dermis. Skin affected by AD oozes before becoming chronically dry and scaly; inflammation associated with CD typically resolves within days.

Although no cures exist for either AD or CD, management of skin inflammation can be achieved through treatments that suppress inflammation. Multiple factors are thought to contribute to the origin of inflammatory skin diseases, including genetic, immune, and environmental causes. Inflammatory skin disease tends to run in families, a phenomenon that may be related to protein defects. For example, an inherited defect in a structural protein synthesized by keratin-producing epithelial cells is thought to be a key driver of AD.

Some research suggests that inflammatory skin disease may be related to a hyperactive immune response. Normally, immune cells discriminate between harmful pathogens and harmless substances. If a harmless molecule that commonly touches the skin causes a response as though the molecule were harmful, further contact may produce an inflammatory response.

The hypothesis that modern hygiene plays a role in the development of inflammatory skin disease has also been advanced. For example, frequent hand-washing with soap and water may remove skin oils, leaving the skin more sensitive to common irritants.

Scientists have developed an experimental treatment for inflammatory skin diseases that modifies the activity of certain T-lymphocytes. One function of T-lymphocytes is to:

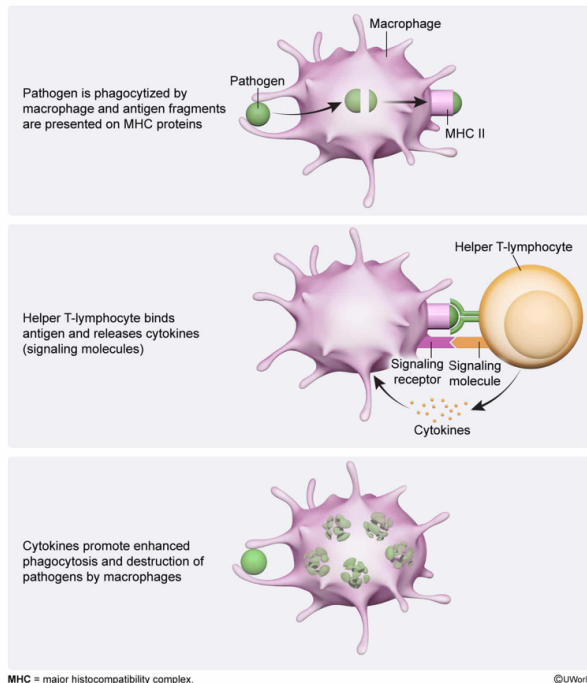
- ☐ A. present foreign antigen fragments on cell membranes. [14%]
- ☐ B. promote a distinct innate immune system response to a specific pathogen. [19%]
- ✓ ☒ C. secrete signaling molecules that activate other immune cells. [56%]
- ☐ D. engulf foreign substances and cellular debris. [8%]

Correct

56% answered correctly

Explanation:

T-lymphocytes secrete signaling molecules that activate other immune cells



The **adaptive immune system** consists of two types of lymphocytes: **B-lymphocytes** and **T-lymphocytes**. These types of cells recognize specific foreign antigens and are able to engage specialized immune responses against pathogens.

T-lymphocytes bind foreign antigens found on cell surfaces and include:

- **Helper T-lymphocytes** produce signaling molecules called **cytokines** that activate other cells of the immune system and enhance immune cell activity.
- **Cytotoxic T-lymphocytes** cause **apoptosis** to occur in pathogen-infected cells.

In this question, scientists have developed an experimental treatment for inflammatory skin diseases (eg, AD, CD) that modifies T-lymphocyte activity. In these types of diseases, a heightened immune response is mounted against allergens or irritants. Therefore, one T-lymphocyte function that may be modified by this treatment is the **secretion of signaling molecules** (ie, cytokines) that **activate other immune cells**. Presumably, this treatment would *decrease* inflammation by activating *fewer* immune cells and returning the immune response to homeostasis levels.

(Choices A and D) **Phagocytic** cells (eg, **macrophages**, dendritic cells), *not* T-lymphocytes, present antigen fragments on cell membranes and engulf foreign substances and cellular debris.

(Choice B) The **innate immune system** offers *nonspecific* protection against a wide variety of invaders. The specific response to antigens by T-lymphocytes is part of the adaptive (*not* innate) immune system.

Educational objective:

T-lymphocytes recognize and bind foreign antigens found on cell surfaces. Types of T-lymphocytes include helper T-lymphocytes and cytotoxic T-lymphocytes.

Time spent:0 sec

Subject:
Biology

QID:401753

System:
Skin and Immune Systems

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The hypothesis that modern hygiene plays a role in the development of inflammatory skin disease has also been advanced. For example, frequent hand-washing with soap and water may remove skin oils, leaving the skin more sensitive to common irritants.

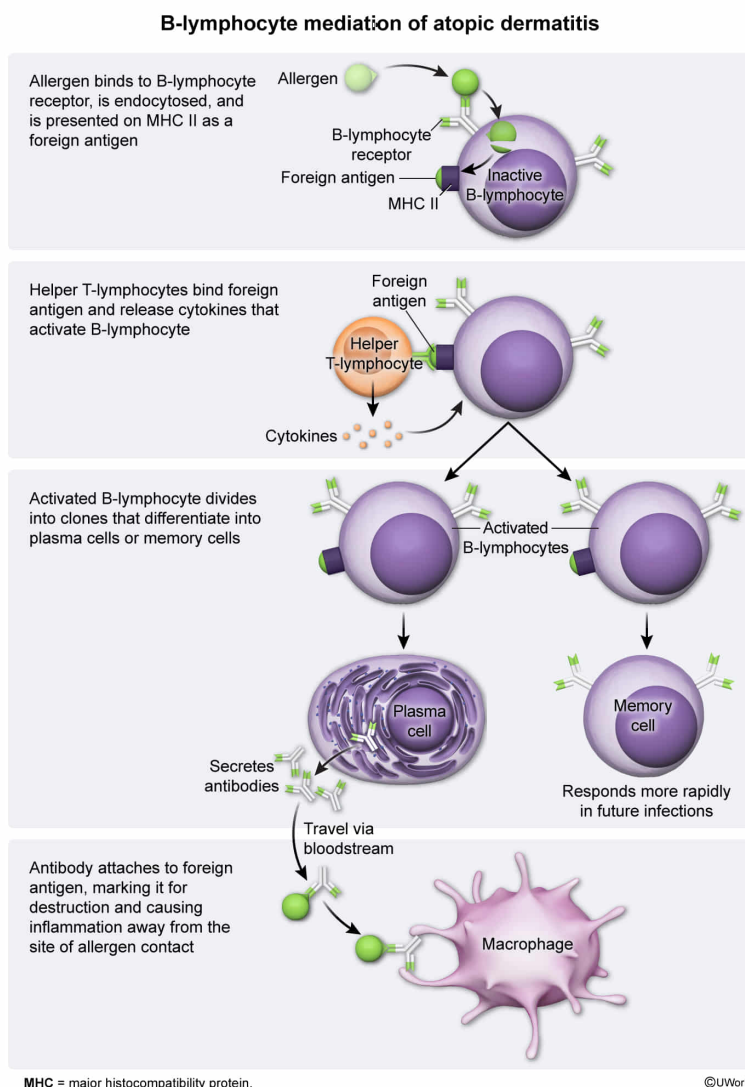
The information in the passage suggests that which characteristic of atopic dermatitis is mediated by B-lymphocytes?

- ☐ A. Joint pain [13%]
- ☐ B. Prevalence within families [12%]
- ☐ C. Involvement of both the epidermis and dermis [6%]
- ✓ ☒ D. Inflammation away from the site of allergen contact [67%]

Correct

68% answered correctly

Explanation:



B-lymphocytes are a type of white blood cell involved in the

adaptive immune system. These cells can recognize specific foreign antigens and mount responses against pathogens. To do this, B-lymphocytes secrete **antibodies** that circulate throughout the blood and lymph. These antibodies bind foreign antigens and mark them for destruction by other immune cells.

B-lymphocytes remain inactive until their cell surface receptors bind a specific foreign antigen, which activates the lymphocyte. Subsequently, the B-lymphocyte takes up the antigen via endocytosis and breaks it into smaller fragments. These fragments are then bound by specialized proteins called major histocompatibility class II (MHC II) proteins and presented on the surface of the B-lymphocyte.

Subsequently, other immune cells can bind the foreign antigen, releasing **cytokines** (signaling molecules) that stimulate the proliferation of B-lymphocytes. These new B-lymphocytes fall into two categories: short-lived **plasma cells** that secrete antibodies, and long-lived **memory B-lymphocytes** that can recognize the same foreign antigen more rapidly in the event of a future encounter by the immune system.

The passage states that atopic dermatitis (AD) commonly affects the skin covering several joints, both exposed to the environmental allergens as well as covered by clothing. This suggests that skin that did not come into contact with an allergen can experience inflammation. Therefore, **inflammation** in AD can occur **away from the original site of allergen contact** due to antibody circulation.

(Choice A) AD is a condition affecting skin covering several joints, according to the passage. However, *no* information is presented to suggest that the joints *themselves* are affected.

(Choice B) The tendency of AD to run in families could be due to

genetic factors (eg, an inherited defective structural protein, as mentioned in the passage), shared environmental exposures (which may or may not elicit pathological B-lymphocyte responses), or both. Therefore, the high prevalence of AD within certain families does *not* necessarily suggest that B-lymphocytes are involved.

(Choice C) According to the passage, AD involves *only* the epidermis, not the epidermis *and* the dermis.

Educational objective:

In response to binding foreign antigens, B-lymphocytes divide and differentiate into plasma cells. Plasma cells secrete antibodies, which mark foreign antigens for destruction by other immune cells.

Time spent:0 sec

Subject:

Biology

QID:401754

System:

Skin and Immune Systems

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Some research suggests that inflammatory skin disease may be related to a hyperactive immune response. Normally, immune cells discriminate between harmful pathogens and harmless substances. If a harmless molecule that commonly touches the skin causes a response as though the molecule were harmful, further contact may produce an inflammatory response.

The hypothesis that modern hygiene plays a role in the development of inflammatory skin disease has also been advanced. For example, frequent hand-washing with soap and water may remove skin oils, leaving the skin more sensitive to common irritants.

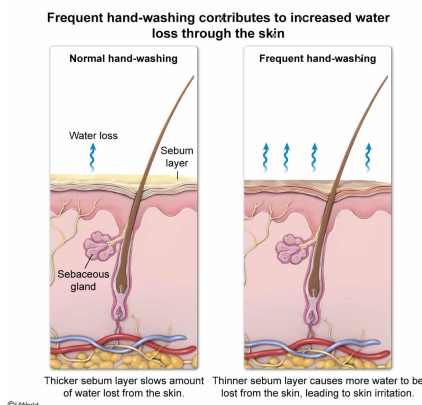
Presumably, frequent hand-washing leads to skin irritation because:

- ☐ A. chemicals used in soap inhibit the function of neutrophils. [25%]
- ☐ B. evaporative cooling of the skin cannot compensate for hand-washing with hot water. [2%]
- ☐ C. soap detergents damage immune cells that promote inflammation. [21%]
- ✔ ☒ D. hand-washing contributes to increased water loss through the skin. [50%]

Correct

50% answered correctly

Explanation:



The **skin** is the body's largest organ and serves several functions:

- Protection from pathogens

- Maintenance of body osmolarity via the prevention of excess water evaporation
- Regulation of **body temperature homeostasis** (ie, thermoregulation)

The passage states that modern hygiene such as frequent hand-washing may remove skin oils and leave skin more sensitive to common irritants. Skin oils (ie, sebum) have a bactericidal function: They are part of the body's **innate immune system** and kill harmful bacteria on the skin's surface.

In addition, these oils slow the amount of water lost from the skin via evaporation. If skin oils decrease, more water is able to evaporate through the skin surface. Therefore, because **frequent hand-washing causes more water to be lost through the skin**, the skin can become more irritated.

(Choice A) Neutrophils, a type of **leukocyte**, function to **phagocytize** and kill bacterial cells as part of the innate (nonspecific) immune system. Neutrophils are found in the circulation but not in the skin unless the skin is wounded. Therefore, chemicals used in soap would likely *not* inhibit the function of neutrophils.

(Choice B) Evaporative cooling is a thermoregulatory response to hot environments. When water is secreted or applied to the surface of the skin and evaporates, the body is cooled. The localized and temporary application of hot water to the hands during hand-washing is *not* an example of a generalized hot environment (ie, surrounding the whole organism) that would provoke significant evaporative cooling mechanisms.

(Choice C) The skin creates a barrier that provides protection from the entry of pathogens. Therefore, soap detergents likely do *not* cross this barrier to damage immune cells.

Educational objective:

Skin has several functions, including protection from pathogens, maintenance of body osmolarity, and thermoregulation.

Time spent: 0 sec
Subject:
Biology

QID: 401755
System:
Skin and Immune Systems

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The hypothesis that modern hygiene plays a role in the development of inflammatory skin disease has also been advanced. For example, frequent hand-washing with soap and water may remove skin oils, leaving the skin more sensitive to common irritants.

Based on the passage, which of the following treatments could best prevent inflammatory skin diseases from recurring?

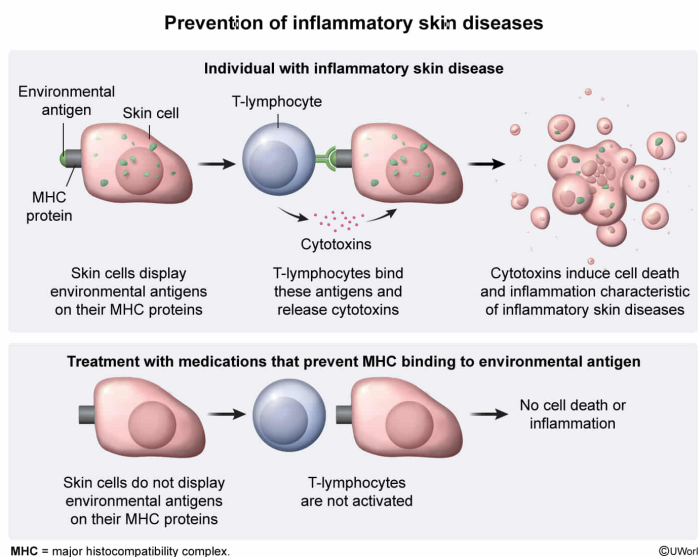
- ☐ A. Use of medications that increase the stability of antibody-antigen complexes [9%]

- ✓ ☒ B. Use of medications that prevent MHC binding to common environmental antigens [79%]
- ☐ C. Use of medications that inhibit the colonization of skin by bacteria [3%]
- ☐ D. Use of medications that inhibit corticosteroid synthesis and secretion [7%]

Correct

78% answered correctly

Explanation:



The T cells (T-lymphocytes) of the **adaptive immune system** recognize harmful antigens (ie, molecules to which the immune system can react) displayed by specialized cell surface proteins known as **major histocompatibility complex (MHC) proteins**. MHC proteins are found on the surface of many types of cells (eg, other **immune cells**, infected cells) and display fragments of proteins present within a given cell to T-lymphocytes. By binding an antigen and the MHC protein that is displaying that antigen, T-lymphocytes are activated and can proliferate.

According to the passage, inflammatory skin diseases such as AD and CD can be managed through treatments that suppress inflammation. To best prevent inflammatory skin diseases from recurring, medications that **prevent MHC binding to common environmental antigens** (ie, those that would cause an inflammatory response) should be used. By

preventing MHC binding, T-lymphocytes will remain inactive, and the inflammatory response likely will be prevented.

(Choice A) Increasing the stability of antibody-antigen complexes would trigger activation of the complement system, a group of proteins that bind nonspecifically to foreign cells and enhance (*not* suppress) inflammation. Therefore, this approach would *not* best prevent inflammatory skin diseases from recurring.

(Choice C) As long as the epidermis is unbroken, the skin serves as a barrier to the entry of pathogens (including bacteria) into the body. Consequently, the presence of bacteria on skin would *not* normally stimulate an inflammatory response. As such, the use of medications that inhibit the colonization of skin by bacteria likely would *not* influence inflammation.

(Choice D) Corticosteroids are anti-inflammatory molecules. Therefore, medications that inhibit corticosteroid synthesis and secretion would increase, *not* decrease, inflammation.

Educational objective:

Major histocompatibility complex (MHC) proteins, or cell surface antigen-presenting proteins, present antigens to which T-lymphocytes can bind. Due to this binding, T-lymphocytes are activated and can proliferate, stimulating the immune response.

Time spent:0 sec

Subject:

Biology

QID:401756

System:

Skin and Immune Systems

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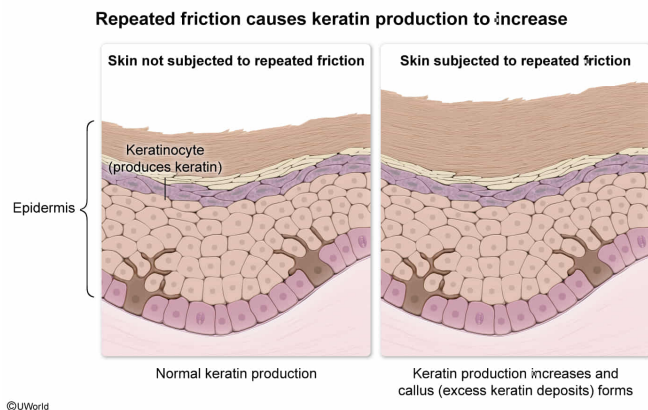
Inflammatory skin diseases affect cells that produce keratin. Under what condition would the level of keratin production likely increase?

- ☐ A. When a drug that inhibits protein synthesis is administered [6%]
- ☒ B. When an area of skin is exposed to repeated episodes of friction [52%]
- ☐ C. When an inflammatory skin disease destroys sebaceous glands [7%]
- ☐ D. When a fingernail has been shortened by clipping [32%]

Correct

52% answered correctly

Explanation:



Specialized cells in the **skin** (**keratinocytes**) make **keratin**, a unique **protein**. Keratin provides protection against injury to the skin. It also creates a "waterproofing" quality to the skin and helps guard against pathogen entry. When the same area of the skin is subjected to repeated friction, **calluses**, or excessive keratin deposits, can form.

This question asks for a condition that could cause the level of keratin production to increase. When an **area of skin** is exposed to **repeated episodes of friction**, extra keratin is produced and deposited to help protect this area from injury.

(Choice A) Keratin is a protein; therefore, administration of a drug that inhibits **protein synthesis** likely would decrease (*not* increase) the level of keratin production.

(Choice C) Sebaceous glands produce sebum, *not* keratin. Therefore, an inflammatory skin disease that destroys sebaceous glands likely would not affect the level of *keratin* production.

(Choice D) Nails are hard, protective coverings at the distal portions of the fingers and toes. Each nail consists of hard keratinized cells derived from the **epidermis**, and each grows in length as new nail cells become keratinized. Trimming portions of the nail that are already keratinized does *not* affect the consistent level of keratin production.

Educational objective:

Keratin, a unique protein produced by specialized cells in the skin, provides

protection against injury and a "waterproofing" quality, and guards against pathogen entry through the skin.

Time spent:0 sec

Subject:
Biology

QID:401757

System:
Skin and Immune Systems

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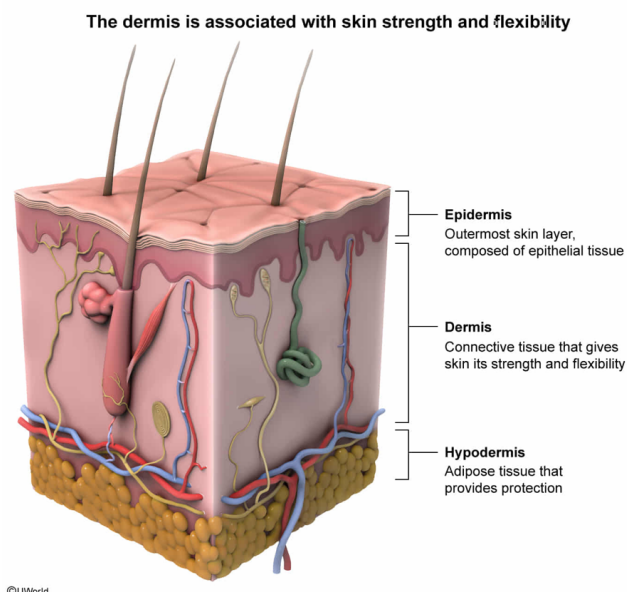
According to the passage, contact dermatitis could cause:

- ☐ A. inflammation of subcutaneous fat and loose connective tissue. [13%]
- ☐ B. damage only to cutaneous tissues derived from ectoderm. [20%]
- ☒ C. damage to the skin layer associated with skin strength and flexibility. [61%]
- ☐ D. chronic vasoconstriction of cutaneous blood vessels. [4%]

Correct

62% answered correctly

Explanation:



The **skin** is divided into three major layers:

1. The **epidermis** makes up the outermost (superficial) layer of skin and is itself divided into five additional layers (strata). It is composed of **epithelial tissue** and is continuously sloughed off (superficial layer) and reproduced (deep layer). Merkel cells, Langerhans cells, melanocytes, and keratinocytes are all cell types found in the epidermis.
2. The dermis extends deep and immediately beneath the epidermis. This layer is composed of a matrix of **connective tissue** embedding several cell types. Blood vessels, hair follicles, sensory receptors, and sweat (sudoriferous) and oil (sebaceous) glands are found in the dermis. The more superficial papillary dermis contributes to skin flexibility, and the deeper reticular dermis contributes to skin strength.
3. The hypodermis (subcutaneous layer) makes up the deepest layer. It is composed primarily of adipose tissue, providing protection for the internal organs by acting as both a thermal insulator and a shock absorber.

The passage states that contact dermatitis (CD) is an inflammatory skin disease usually confined to skin that comes into contact with allergens or irritants, which would be the epidermis. However, the passage also states that CD may involve both the epidermis and the dermis. Because the dermis is the **skin layer** associated with **skin strength and flexibility**, CD could therefore

cause damage to this skin layer.

(Choice A) The hypodermis contains subcutaneous fat and loose connective tissue. The passage states that CD affects the epidermis and dermis, *not* the hypodermis.

(Choice B) The **ectoderm**, or the outermost germ layer of an embryo, gives rise to the epidermis. However, the mesoderm (middle germ layer of an embryo) gives rise to the dermis, which the passage states can also be affected by CD. Therefore, CD would not cause damage to cutaneous tissues derived from ectoderm only, but rather tissues derived from *both* ectoderm and mesoderm.

(Choice D) During an inflammatory response such as that caused by CD, blood vessels **dilate** and become more permeable. This allows increased blood flow, bringing more immune cells and chemicals to the affected area. Therefore, CD could cause chronic vasodilation (*not* vasoconstriction) of cutaneous blood vessels.

Educational objective:

The skin is divided into major layers: the epidermis, dermis, and hypodermis (subcutaneous layer), moving from superficial to deep. The epidermis is subdivided into five additional layers and is composed of epithelial tissue. The dermis is connective tissue that contributes to flexibility and the strength of the skin. The hypodermis (subcutaneous layer) protects internal organs.

Time spent:0 sec

Subject:

Biology

QID:401758

System:

Skin and Immune Systems